Preliminary Datasheet

Ultramid®

A 60G1 V30 OR16655

01/2024

PA66-GF30 FR(40)

Product description

Without halogens flame retardant glass fiber reinforced injection moulding grade. Ultramid® A 60G1 V30 OR16655 is orange color and provides good mechanical and electrical properties. Due to the halide free stabilization the impact on corrosion is minimized and sensitive electronic components are better protected.

Physical form and storage

The product is supplied in the form of granules with a bulk density of approx. 0.7 g/cm³. Standard packs are bag and bulk container (octagonal IBC=intermediate bulk container made from corrugated board with a liner bag). Other packaging materials and shipping in road or rail silo wagons are possible by agreement. The containers should only be opened immediately before processing or drying. To ensure that the delivered product absorbs as little moisture as possible, the containers should be stored in dry rooms and always carefully closed again after partial quantities have been withdrawn. In principle, the product can be stored for a long period of time. Containers stored in cold rooms should be equalized to ambient temperature before opening in order to avoid condensation on the granules. Regardless of the storage conditions, the product should be pre-dried according to our recommendations and the machine should preferably be loaded using a closed conveyor system.

Product safety

In case processing is done under conditions as recommended (cf. processing data sheet) melts are thermally stable and do not generate hazards by molecular degradation or the evolution of gases and vapors. Like all thermoplastic polymers the product decomposes on exposure to excessive thermal load, e.g. when it is overheated or as a result of cleaning by burning off. Further information is available from the safety data sheet.

Note

The data contained in this publication are based on our current knowledge and experience. In view of the many factors that may affect processing and application of our product, these data do not relieve processors from carrying out their own investigations and tests; neither do these data imply any guarantee of certain properties, nor the suitability of the product for a specific purpose. Any descriptions, drawings, photographs, data, proportions, weights etc. given herein may change without prior information and do not constitute the agreed contractual quality of the product. It is the responsibility of the recipient of our products to ensure that any proprietary rights and existing laws and legislation are observed.
In order to check the availability of products please contact us or our sales agency.

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Preliminary Datasheet[3)]

Typical values at 23 °C[1)]	Test method	Unit	Values[2)]
Properties			
Polymer abbreviation	-	-	**PA66-GF30 FR(40)**
Density	ISO 1183	kg/m³	**1440**
Water absorption, 24 h in water, 23 °C	ISO 62	%	**0.73**
Processing			
Melting temperature, DSC	ISO 11357-1/-3	°C	**261**
Melt temperature, injection moulding/extrusion	-	°C	**265 - 300**
Mould temperature, injection moulding	-	°C	**60 - 90**
Molding shrinkage (parallel)	ISO 294-4	%	**0.35**
Molding shrinkage (normal)	ISO 294-4	%	**1.00**
Thermal properties			
Deflection temp. under load 1.8 MPa (HDT A)	ISO 75-1/-2	°C	**248**
Deflection temp. under load 1.8 MPa (HDT A)	ISO 75-1/-2	°C	**261**
Flammability			
UL 94 rating (thickness)	UL-94, IEC 60695	class (mm)	**V-0 (0.8)**
UL 94 rating (thickness)	UL-94, IEC 60695	class (mm)	**V-0 (1.6)**
GWFI (thickness)	IEC 60695-2-12	°C (mm)	**960 (0.8)**
GWIT	IEC 60695-2-13	°C	**775**
thickness	IEC 60695-2-13	mm	**0.8**
Electrical properties			**dry / cond.**
Volume resistivity	IEC 62631-3-1	Ohm*m	**1E14 / -**
Electric strength (d=0.8mm)	IEC 60243-1	kV/mm	**38 / -**
CTI, solution A	IEC 60112	-	**600 / -**
Mechanical properties			**dry / cond.**
Tensile modulus	ISO 527-1/-2	MPa	**11500 / -**
Stress at break	ISO 527-1/-2	MPa	**166 / -**
Strain at break	ISO 527-1/-2	%	**2.6 / -**
Flexural modulus	ISO 178	MPa	**11000/ -**
Flexural strength	ISO 178	MPa	**244 / -**
Charpy unnotched impact strength, 23°C	ISO 179/1eU	kJ/m²	**63 / -**
Charpy notched impact strength, 23°C	ISO 179/1eA	kJ/m²	**9 / -**

Footnotes
1) If product name or properties don't state otherwise.
2) The asterisk symbol '*' signifies inapplicable properties.
3) The typical values of preliminary datasheets are not statistically firm.

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Internal